

Architecture Justification

Web-Based ML Model Evaluator

Selected Architecture: Layered / Three-Tier Architecture

The system is organized into a Presentation Layer, Application/Business Logic Layer, and Data/Model Layer. This architecture separates user interaction from ML evaluation processing and from model/dataset storage.

Reason 1 — Separation of Concerns

Each layer has a focused responsibility. The user interface handles interaction, the application layer performs validation and evaluation, and the data/model layer manages ML models and datasets. This reduces coupling and makes individual components easier to modify and test.

Reason 2 — Maintainability and Extensibility

The architecture allows new evaluation metrics, model formats, datasets, or interface features to be added without redesigning the whole system. Changes in one layer can generally be isolated from the others.

Security Advantage

Input and file validation can be centralized in the application layer before a model or dataset is processed. This provides a clear control point for checking file types, required inputs, and invalid requests.

Performance Advantage

Separating the evaluation service from the presentation layer allows computational ML evaluation work to be handled independently of the user interface. The evaluation component can therefore be optimized or scaled without changing the interface.

Conclusion

The layered architecture is appropriate for the Web-Based ML Model Evaluator because it provides clear separation, easier maintenance, controlled validation, and room for future expansion.